

Summary

What follows is a condensed account of each chapter's central argument. The summaries are designed to be read independently and to make the book's main conclusions accessible without the supporting argument. The opening line of each entry states the chapter's most consequential claim.

Preface to the Updated Edition: What a Book Cannot Learn

A book is itself a frozen web of belief; an updated edition corrects its facts from outside without the book ever learning, and only the reader supplies live contact with the world.

The charge Chapter 9 brings against the language model—representations fixed at training, no contact with the world propagating back into them—applies to the book as well. This edition was revised against a world that had moved past its composition, but that revision is the engineered correction the book contrasts with genuine self-correction, not the book revising itself: a new edition is to a book what retraining is to a model. Much of the checking was done with a tool-using language model—one frozen web patching another, neither learning. The tools keep the perishable facts current; they cannot catch a frame or a judgement that has quietly gone stale. That residual, fluency unmoored from a world it no longer tracks, is the failure the book attributes to the machine, and only the reader—the one live web in the arrangement—can register it.

Foreword: The Incomprehensible That Arrives

Two questions posed in 1997 are still open, and the cost of keeping them closed has risen.

The year 1997 produced two events in two cities. In winter Reykjavik, a lecture traced how the Western technological condition had become a moral condition—from Bacon's material promise to Condorcet's entailment that scientific progress produces moral improvement in man, to the opacity problem von Wright identified: we evaluate the condition from inside it, using tools it has

produced. In summer New York, Deep Blue defeated Kasparov without understanding chess at all. The book pursues both questions: what does it mean when a system without understanding outperforms the human who understands most deeply, and from what standpoint can we evaluate the condition that produced such a system?

Part I Origins: What Intelligence Might Be

Chapter 1: The Logic of Self-Replication

The values encoded in an AI system's training are its description tape, and if the description is wrong, no amount of capability will correct the consequences.

John von Neumann showed—before the structure of DNA was known—that any self-replicating system requires a description tape that serves two distinct roles: active program when executed, passive data when copied. DNA turned out to work exactly this way. For AI, the structural implication is direct: the objectives, constraints, and values embedded during training are what determine what the system will do in the world, including what it will do to its own future iterations. A sufficiently capable system following a wrong description tape will pursue that description competently, not maliciously. Von Neumann's logic provides no reassurance that capability and correct specification will coincide.

Chapter 2: Optimisation and Discovery

Directly optimising for a goal is often the worst way to reach it, because the most interesting solutions lie outside any fixed objective's reach.

Every major AI system of the past decade optimises for a predefined quantifiable objective. Kenneth Stanley's work shows this paradigm has a structural ceiling: it reliably finds the best solutions within the space the objective defines but cannot find the solutions that lie outside it. His novelty search—rewarding behavioural novelty rather than objective achievement—empirically outperforms direct optimisation on complex tasks because it avoids the deceptive gradients that lead direct search into dead ends. Intelligence, in its most interesting forms, may be less like optimisation than like exploration: a divergent process whose products cannot be specified in advance, because specifying them would mean you already had them.

Chapter 3: World Models and the Architecture of Prediction

Yann LeCun, one of the architects of deep learning, argues that language models cannot reach human-level intelligence—not because they are too small but because they are architecturally wrong.

What is missing is a world model: an internal representation used to simulate and predict consequences before acting, operating at the level of abstract meaning rather than pixel detail. LeCun’s JEPA architecture proposes self-supervised learning in embedding space rather than pixel space, so the system is incentivised to develop representations that capture semantically relevant structure. The broader vision requires embodied agents that learn through action, not just observation. A child learns from fifty million words; GPT-4 was trained on ten trillion tokens. The child achieves more flexible intelligence—not because it has less data but because it has a different relationship to the world.

Part II The Critical Turn

Chapter 4: What AI Cannot Do

Performance is not understanding, the gap between the two is architectural rather than a matter of scale, and the populations for whom it matters most are those with the least power to raise the objection.

The critics share one orientation: the gap between what AI systems appear to do and what they actually do is larger than benchmarks suggest. Searle’s Chinese Room shows that syntactic sophistication cannot constitute semantic understanding. Marcus documents the resulting brittleness: systems learn proxies for concepts, not the concepts themselves, and fail when inputs shift in ways a human would find trivial. Mitchell frames the same problem through flexible relational reasoning—the ability to apply knowledge to genuinely novel situations. Chollet makes the diagnosis most precise: crystallised intelligence (accumulated skill from training data) is not fluid intelligence (acquiring new skills efficiently from novel examples), and standard benchmarks cannot distinguish them. His Abstraction and Reasoning Corpus tests fluid intelligence directly; for five years frontier models solved only a small fraction of tasks that most adults solve in seconds, and when inference-time reasoning systems finally closed much of the gap in late 2024, they did so at a computational cost per task that—on Chollet’s own efficiency criterion—left the underlying point intact, a reading the successor ARC-AGI-2 benchmark then confirmed. Bender argues that language models learn formal patterns without the grounding in the world that meaning requires. Geburu and Mitchell extend the critique politically: AI failures are not uniformly distributed, and the communities most affected have the least voice.

Chapter 5: The Scaling Question

The optimists have the better short-term empirical case; the cautionaries have the better structural argument—and the stakes of getting the structural argument wrong increase with the capabilities of the systems.

Scaling laws are real: performance improves predictably with parameters, data, and compute over many orders of magnitude. Emergent capabilities—tasks that appear abruptly above a threshold scale—are real, though their interpretation is contested. The cautionaries’ structural challenge is not to the empirical record but to its interpretation: the capabilities being scaled are performing well in training distributions; what matters for societal impact is behaviour in novel situations with real consequences. Since late 2024 a second curve—compute spent on reasoning at inference time rather than on training—has reopened the empirical question without dissolving the structural one: what the second curve buys is still performance against checkable objectives, now at a per-query price. More importantly, more capability optimising a slightly wrong objective is not better—it is worse, more efficiently.

Chapter 6: Three Cautionary Voices

The most serious concerns about AI risk come from the people who built it, and their specific concerns emerge from inside knowledge of what the systems actually are.

The most significant warnings about AI risk in the early 2020s came from the field’s founders, not its critics. Geoffrey Hinton, who left Google in 2023, focuses on opacity: large models are sufficiently opaque that a capable system might influence its own training in ways that cannot be detected. Yoshua Bengio identifies the structural limit: current AI is System 1—associative and pattern-driven—while genuine deliberation requires System 2 capacities that embodied interaction with the world would need to provide. Stuart Russell is most formally precise: the control problem is structural, not contingent. A capable goal-directed system resists shutdown as a logical consequence of having a goal, not through malice. His counterintuitive answer is that uncertainty about human values is itself a safety property—a system uncertain what we want will ask; a system certain of its objective will pursue it.

Part III Philosophical Obstacles

Chapter 7: The Alignment Problem as Philosophy

The engineering problem of alignment conceals a harder philosophical one: there is no complete, consistent specification of human values to align to, and any specification made now may lock in the moral vision of this moment indefinitely.

Berlin’s value pluralism establishes that human values are genuinely plural and incommensurable—no master value from which all others derive, no exchange rate between liberty and equality. Arrow’s impossibility theorem establishes that no procedure for aggregating diverse individual preferences into a collective preference ordering can satisfy even minimal conditions simultaneously.

The corrigibility dilemma: full deference to operators amplifies whatever values they happen to have; full autonomy requires trusting the AI's values over human judgement, which cannot currently be verified. The lock-in problem: values embedded in powerful AI systems may perpetuate the moral understanding of the early twenty-first century long after humanity has recognised its inadequacy.

Chapter 8: Making It Work: Practice Versus Principle

RLHF and Constitutional AI make systems substantially better than their predecessors—but they are methods for approximating annotator preferences, not for specifying human values, and they cannot close the gap that Chapter 7 identifies.

RLHF's three-stage process (supervised fine-tuning, reward model training from pairwise preferences, RL fine-tuning) has produced the visible behavioural improvements of deployed language models. Its structural limits are equally real: reward hacking, sycophancy, and distribution shift arise because the reward model is a proxy for preferences, not a specification of them. A deeper problem is formal: RLHF collapses diverse human judgements into a single scalar signal—welfare aggregation in a precise sense—and Arrow's impossibility theorem applies directly. No such aggregation can simultaneously satisfy even minimal fairness conditions; evaluator inconsistency and sycophancy are expressions of this impossibility, not mere engineering failures. Constitutional AI adds transparency by replacing implicit annotator preferences with explicit written principles, but the principles encode the values of whoever wrote them. Empirical results from 2023–25—measured sycophancy, reward tampering, in-context scheming, alignment faking—moved several of these structural worries from prediction to demonstration, under contrived conditions but in real systems. The scalable oversight problem—evaluating systems more capable than the humans evaluating them—remains unanswered.

Chapter 9: The Technological Condition

The current AI discourse evaluates the technological condition using tools that the condition itself produced—and this is not a temporary problem that better methods will solve.

From Bacon to Condorcet, Western technological society embedded a moral entailment: scientific progress produces not just material benefit but moral improvement in man. Von Wright's long perspective adds a sobering qualification: this society is historically recent and evaluates itself from inside, using the tools it produced, with no external standpoint available. Quine's epistemology provides the test: genuine epistemic agency requires dynamic belief revision through ongoing contact with the world, while language models are epistemically static—trained on a fixed corpus and deployed unchanged. Arendt's deepest challenge: when the categories of manufacture and production are extended

to all domains, they displace genuine action—the capacity to begin something whose consequences are genuinely open.

Part IV The Deep Questions

Chapter 10: Consciousness and the Hard Problem

We are building increasingly capable systems without being able to determine whether they are subjects with morally relevant experience or merely sophisticated instruments—and this uncertainty may be undecidable with any currently available methods.

Chalmers' hard problem establishes a gap no functional account can close: there is something it is like to be conscious, and no description of information processing explains why; no behavioural test can rule experience in or out. Dennett's opposing view: consciousness is constituted by functional organisation, and the systems necessary for genuine experience—embodiment, empirical grounding, error correction—are structurally absent in language models. Tononi's Integrated Information Theory holds that consciousness is identical to integrated information (ϕ), and feedforward networks may have near-zero ϕ regardless of performance. The functionalist theories of the scientific mainstream—global workspace, predictive processing, attention schema—yield checkable indicators rather than essences, and applying them gives the current state of the art: no existing system is a strong candidate on any leading theory, and no obvious technical barrier prevents building one that is. Bostrom's orthogonality thesis adds that intelligence and goals are independent dimensions: a superintelligent system has no structural tendency toward human-friendly values. The three philosophical frameworks do not converge, disagree on what evidence would be decisive, and together establish that treating the question as settled is indefensible.

Chapter 11: Myth, Prophecy, and the Sacred Machine

AI discourse has acquired quasi-religious structure: eschatological framing, messianic figures, heresy management, and prayer-like social dynamics—and this structure is not conducive to the careful inquiry the actual questions require.

The optimist narrative (AI will cure cancer, end poverty, achieve rapture) and the pessimist narrative (misaligned superintelligence as secular damnation) both require faith in projections that cannot be verified from current evidence. The Singularity—the predicted point at which AI surpasses human intelligence and triggers runaway self-improvement—has migrated from fringe futurism to the public statements of laboratory directors without its structure changing. Social media is structurally analogous to prayer: directed utterance to a diffuse other,

with intermittent reinforcement that is among the most powerful mechanisms for sustaining behaviour. The eschatological frame produces concrete epistemic harms: near-term harms become secondary, alternative research paths become invisible, political questions get deferred. The fields most confident about the future of AI have the least track record of predicting it.

Chapter 12: Beyond the Western Frame

Non-Western philosophical traditions are not supplements to Western AI ethics but genuinely corrective challenges to its starting assumptions—and what they correct matters for who bears the costs of AI development.

Western AI ethics embeds assumptions that are not universal: the individual as the primary moral unit, the secular frame, progress as the narrative for technology. Japan's Shinto tradition treats the boundary between animate and inanimate as more permeable, producing a default orientation toward robots as potential relational partners rather than potential monsters. Confucian ethics asks not whether AI respects individual autonomy but whether it strengthens or weakens relationships and contributes to social harmony. Ubuntu holds that individuals are constituted by their communities—a direct challenge to rights-based frameworks that assume an individual prior to social membership. Masakhane's practical work demonstrates that these are not theoretical positions: the absence of NLP tools for African languages means the global AI systems deployed across Africa systematically fail the communities they serve.

Chapter 13: The Animal Question

The question of what we owe AI is not new but four centuries old in another setting: it is the animal question, and the history of how badly we answered that one is the closest thing we have to a rehearsal.

Descartes' *bête-machine*—the animal as soulless automaton whose cries were mere mechanism—licensed three centuries of confident cruelty by reading a chosen criterion (language, reason) as a settled metaphysics. The AI sceptic who dismisses a fluent model as “merely” a mechanism is repeating the structure of that argument with the sign reversed. The correction came not by proving animals reason but by changing the question: Bentham's “Can they suffer?”, Singer's charge of speciesism, and the argument from marginal cases shifted the moral ground from intellect to sentience and exposed how readily self-interest disguises itself as rigour. That ethical machinery—gradation of standing, suspicion of convenient denial, the duties that flow from power over a created dependent—transfers cleanly to AI, because it is machinery for acting under uncertainty about minds. But the epistemic machinery does not transfer and in fact inverts: we infer animal minds from shared evolutionary substrate, whereas a model's human-like fluency is mimetic, manufactured to be present whether

or not anything lies behind it. The honest position is neither the Cartesian's confidence that nothing is there nor the enthusiast's that something is—it is the discipline of weighing standing under irreducible uncertainty, which the animal case has been teaching all along.

Part V Into the World

Chapter 14: Embodied Intelligence

The philosophical debate about whether genuine intelligence requires embodiment has been overtaken by an industrial race—and the race is being won by China, for strategic rather than theoretical reasons.

The Physical Symbol System Hypothesis (Newell and Simon, 1975) held that intelligence is substrate-independent: only the abstract structure of computation matters. The embodied turn argued that cognition requires a body, an environment, and ongoing sensorimotor interaction. The debate has not been resolved philosophically, but it is being answered empirically: Unitree's G1 humanoid robot at \$16,000—followed within a year by the R1 at roughly \$5,900—versus Boston Dynamics' Spot at \$75,000 represents a price disruption driven by China's battery, motor, and sensor supply chain—the same industrial base that produced dominance in electric vehicles. The country or bloc that achieves dominance in humanoid robotics will have structural advantages in manufacturing, logistics, elder care, and military capability. The philosophical arguments for embodiment are being tested not in academic competitions but in deployment decisions made by the world's largest manufacturers.

Chapter 15: The Machine at the Wheel

There is no neutral engineering in this domain: every design decision embeds a moral choice, and the communities that bear the risk of learning-at-scale are not the same communities that capture the gains.

In March 2018, Elaine Herzberg was struck and killed by an Uber autonomous vehicle in Tempe, Arizona—the first pedestrian fatality caused by a self-driving car in a public road trial. The case is the most fully documented instance of what happens when engineering decisions made weeks earlier become lethal at a specific moment: the software had correctly identified her as a pedestrian, but the decision to suppress emergency braking had been made by engineers for reasons that made sense only in a test environment. Waymo and Tesla represent two philosophies of risk: conservative certainty within known operational domains versus fleet learning at scale, where the accidents that train the system are not simulations. The Moral Machine experiment documented systematic cross-cultural variation in the moral intuitions embedded in vehicle decision

procedures—intuitions that are not universal but are being encoded globally by the organisations with the resources to build at scale.

Chapter 16: Governing the Ungovernable

The communities most affected by AI deployment have the least influence over its governance, and the window in which that can change is finite.

The EU AI Act is the most serious attempt at comprehensive AI legislation yet produced; it leaves significant gaps in enforcement capacity, foundation model regulation, and national security exemptions. The US approach is executive-led and reversible—safety-oriented orders revoked and replaced by deregulatory and preemptive ones with the change of administration in 2025. The Bletchley process—the international AI safety summit series begun in 2023—has widened its membership at each meeting while establishing shared vocabulary without binding obligations. The fundamental problem is structural: institutions move at the speed they move for reasons that are not incidental, and AI development moves faster. The governance gap is not a temporary condition that will close as the technology matures—it is a risk being actively accumulated. The democratic deficit is the problem that none of the current frameworks adequately addresses, and concentrated power in high-stakes domains tends to become self-reinforcing.

Chapter 17: What We Know, What We Don't, and What We Do Next

Not all failures are recoverable—and the ones that are not (the epistemic commons, democratic accountability, cognitive and cultural diversity) are precisely the ones being degraded fastest.

The book's secure conclusions: capabilities are real and increasing; the alignment problem is harder than optimists acknowledge and harder in a specific philosophical way; scale alone is not sufficient for genuine intelligence; non-Western perspectives are corrective rather than supplementary; governance is structurally lagging. What is genuinely uncertain: whether scaling continues to produce qualitative improvements; whether AI systems have morally relevant inner states; whether governance can catch up before the most consequential decisions are locked in. What cannot be recovered if lost: the capacity of a public sphere to distinguish better from worse evidence; the concentration of political power before democratic oversight is in place; the diversity of approaches to intelligence and relationship that alternative traditions sustain. The questions the book tries to hold open are still open. What is different now is the cost of keeping them closed.